

***In situ* grafting onto solution polymerized styrene butadiene rubbers (SSBR) filled with silica via solid state method**

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ABSTRACT: Grafting reaction of solution polymerized styrene butadiene rubbers (SSBR) and 3-mercaptopropionic acid *in situ* was investigated in a solid state and a series of SSBR with different polarity degree were prepared through grafting 3-mercaptopropionic acid onto SSBR chains in the process of mixing. The feasibility of solid-phase grafting *in situ* (SPGiS) was verified by ¹H-nuclear magnetic resonance and Fourier transform infrared spectroscopy. Compared with unmodified SSBR/silica compounds, the SSBR/silica compounds with SPGiS exhibited preferable performance in silica dispersion, rubber-filler interaction, and reinforcing efficiency. The scanning electron microscope and transmission electron microscope showed that silica presented better dispersion and smaller aggregates in the SSBR of SPGiS. Dynamical mechanical analysis of vulcanized rubber compounds with SPGiS showed that the wet skid resistance and the rolling resistance of SSBR/silica compounds were improved separately. © 2018 Wiley Periodicals, Inc. *J. Appl. Polym. Sci.* **2018**, 135, 46653.

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INTRODUCTION

In 1990s, the so-called “Green tire” by Michelin was introduced in a United States Patent¹ and the patent disclosed a filling technology of precipitated silica for the manufacturing of high-performance tires. Silica as a reinforcement agent provides a combination of good mechanical properties, high resilience, excellent rolling resistance, and low heat build-up. Since then, the successful application of silica in high-performance tires has attracted considerable interest from both industry and academic fields.^{2–4}

SSBR/silica compound as one of the most important materials in green tires provides a combination of good mechanical properties, low rolling resistance, and high wet skid resistance.^{5–8} Because of the presence of numerous silanol groups on the surface of silica, a major problem in the application of silica is its poor dispersion in the rubber matrix, which is caused by the great difference in surface polarity between silica and rubber.^{9,10} However, the dispersion degree of filler and the interaction between filler and rubber are the key factors to determine the performance of compounds.^{11,12} To solve this problem, many

efforts have been devoted by researchers at home and abroad.^{13–16} To date, the most successful achievement is the application of silane coupling agent to enhance the filler dispersion. Although the affinity between rubber macromolecules and silica modified by a silane coupling agent has been improved to some extent, the problem caused by large specific surface area of silica and strong self-aggregation is still not eliminated completely. In recent years, surface modification of silica^{17–21} and the development of new highly dispersed silica^{22–24} have also been subjects of interest. In addition, the polarity modification of rubber is also an effective method to improve the dispersion of silica and enhance the interaction between filler and rubber. Some functional groups, such as —COOH, —OH, and —NH₂ have been introduced into rubber molecules and some exciting results have been obtained in dynamic mechanical properties.^{25–28} In the literature, such research work has mostly focused on the preparation of modified rubbers and characterization of macroscopic properties. The preparation of modified rubbers often has been carried out in solution. The disadvantages of this modified method in solution are high energy consumption, heavy pollution, and low efficiency. However, few

findings have been published in the literature related to the way of SPGIS to improve the dispersion of filler and enhance the interaction between filler and rubber.

In this article, because of the existence of the double bonds in the molecular chain of SSBR, we investigated the possibility of a grafting reaction *in situ* between 3-mercaptopropionic acid and SSBR in solid melting state and investigated the performance of SSBR/silica compounds obtained from SPGIS, including silica dispersion, filler-filler interactions, filler-rubber interactions and the dynamic mechanical property, etc. Grafting effect between 3-mercaptopropionic acid and SSBR was verified in this article through ^1H -nuclear magnetic resonance (NMR) and Fourier transform infrared (FTIR) spectroscopy. The influence of different amounts of 3-mercaptopropionic acid on the performance of SSBR/silica compounds was discussed, which would reveal the relationship between different polarity degree of SSBR and the application performance of SSBR/silica compounds.

EXPERIMENTAL

Materials

Commercial SSBR2506 (styrene content = 23.4 wt %, vinyl content 65.0 wt %, ML (1 + 4) 100 °C = 63) was purchased from Yanshan Petrochemical Co. Sinopec (Beijing, P.R.C.). The modification agent was 3-mercaptopropionic acid (99%, Aldrich). The precipitated silica (Zeosil 1165MP) with a specific surface area of 165 m² g⁻¹ provided by Rohdia. The silane coupling agent, bis-(triethoxysilylpropyl)-tetrasulfide, was obtained from Nanjing Shuguang Chemical Group Co., P.R.C. The curing accelerators were N-tert-butylbenzothiazole-2-sulphenamide (TBBS) (>97%, Puyang Willing Chemicals Co., Henan, P.R.C.) and tetramethylthiuram disulfide (TT) (>97%, Aladdin, Shanghai, P.R.C.).

Preparation Processing

Preparation of modified SSBR *in situ* is introduced as following. The weight proportions of SSBR and 3-mercaptopropionic acid were 100:0.2, 100:0.4, 100:0.8, and 100:1.6, respectively. A mixture of SSBR and 3-mercaptopropionic acid was prepared in an internal mixer (Thermo Fisher Scientific, Rheomix 3010 OS, Banbury rotors) at temperature of 90 °C and rotor speed of 77 rpm for 3 min.

The formulation of the compounds is listed in Table I. Processing method is introduced as following. First, adjust the temperature of the laboratory internal mixer (American Farrel Cor., BR1600, Banbury rotors) to a starting temperature of 90 °C and rotor speed of 77 rpm, SSBR2506 was masticated for 0.5 min, then 3-mercaptopropionic acid was mixed in rubber to ensure the reaction of grafting completely for 3 min, and at last silica, bis-(triethoxysilylpropyl)-tetrasulfide, ZnO, and stearic acid were added into the mixer for 5 min ensure silanization reaction between silane and silica. The curing package consists of 1.5 phr sulfur, 1.5 phr TBBS, and 0.2 phr TT. The cure time was determined using an oscillating disc rheometer (ODR) at 160 °C.

Measurements and Characterization

The structural units of all samples grafted in SSBR, including styrene units and vinyl contents from 1,2-polybutadiene and

Table I. formulation for Preparation of the Compound Samples

Ingredient	Amount (phr) ^a
SSBR	100
3-Mercaptopropionic acid	0.2, 0.4, 0.8, 1.6
Silica	60
Bis-(triethoxysilylpropyl)-tetrasulfide	4.8
ZnO	3.0
Stearic acid	2.0
TBBS ^b	1.5
TT ^c	0.2
Sulfur	1.5

^a The number of rubber, filler, and additives in phr (parts by weight per hundred parts of rubber).

^b N-tert-butylbenzothiazole-2-sulfenamide.

^c Tetramethylthiuram disulfide.

1,4-polybutadiene, were calculated using ^1H -NMR spectroscopy acquired on a Bruker AV 400 MHz NMR spectrometer. Magnetic field intensity is 9.40 tesla and chemical shifts are given relative to a tetramethylsilane internal reference. The working frequency of hydrogen spectrum is 400.13 MHz, spectral width is 8012.882 Hz, data point is 32 K, and the number of sampling times is 16. The integrals in the proton spectra were obtained directly from concentrated solutions in 1,1,1,2-tetrachloroethane- d_2 . It is noted that all SSBR grafted were extracted with ethanol for 24 h to remove the unreacted chemicals and then dried in vacuum at 40 °C for 12 h.

Gel permeation chromatography analyses of molecular weight and its distribution were performed on a Alliance2690 system (Waters) consisting of a 5 μm and 7.5 mm $\times$ 300 mm PLgel Mixed-C column and a refractive-index detector. The eluting solvent was tetrahydrofuran at a flow rate of 1.0 mL min⁻¹. The retention times were calibrated against polystyrene standards.

FTIR analysis of all samples grafted in SSBR was carried out on a TENSOR 27 FTIR spectrometer (BRUKER, German). The samples were pressed into pellets with KBr. The spectral range was 1200–2000 cm⁻¹. A resolution of 4 cm⁻¹ was chosen. It is noted that the sample grafted was extracted by ethanol for 24 h to remove the unreacted chemicals and then dried in vacuum at 40 °C for 12 h.

Gel content analysis was carried out as following: the sample was placed in the hanging basket soaking in a tetrahydrofuran solution and kept from light for 24 h, then sample dissolved in the hanging basket was dripped by tetrahydrofuran three times. The hanging baskets with the samples were dried in vacuum at 120 °C for 1 h, then removed from vacuum, cooled to room temperature, and weighed. Gel constant could be calculated by: $(m_2 - m_1)/m \times 100$; where m is sample quality, g; m_1 is the weight of the hanging basket, g; m_2 is the weight of the hanging basket and the gel, g.

Bound rubber content was determined by extracting the unbound materials, such as nonrubber ingredients and free

rubbers, with toluene for 7 days at room temperature and dried for 2 days at room temperature. Masses of the samples before and after extraction were measured and the bound rubber content was calculated as follows:

$$R_b(\%) = 100 \times \frac{W_{fg} - W_t[m_f/(m_f + m_r)]}{W_t[m_f/(m_f + m_r)]}$$

where R_b is the bound rubber content, W_{fg} the mass of filler and gel, W_t the mass of the sample, m_f the mass fraction of the filler in the compound, and m_r the mass fraction of the rubber in the compound.

The cure characteristics of SSBR/silica compounds of SPGiS were measured on a Gotech Moving Die Rheometer (GT-M2000-A), which was employed to determine the scorch time (tc_{10}), cure time (tc_{90}), and torque difference ($M_H - M_L$), according to ISO 3417. Samples of the respective compounds were tested at 160 °C. The compounds were subsequently compression-molded using a stainless steel mold at 160 °C, with a pressure of 15 MPa, based on respective curing times.

The tensile fractured surfaces of the samples were examined with scanning electron microscopy (SEM) which were observed by QUANTA SEM (Thermo Fisher Scientific, Massachusetts, USA) on gold-coated surfaces. The fractured pieces were coated with a layer of gold to eliminate electrostatic charge build-up during examination. The working voltage was 20.00 kV.

The micromorphologies of the samples were examined with transmission electron microscopy (TEM). Ultrathin sections of the specimens were prepared at −100 °C using a Leica Ultracut-R ultra-microtome. The thin slices were put on copper grids and then submitted to TEM observation with a F30 TEM (FEI Instrument, The Netherlands) under an accelerating voltage of 300 kV.

The rubber-filler interaction was determined by swelling the cured SSBR/silica compounds of SPGiS in toluene, according to ISO 1817. Test pieces with dimensions of 30 mm × 5 mm × 2 mm were prepared from the molded sheets. The initial weights were recorded prior to testing. The test pieces were then immersed in toluene and conditioned at room temperature in a dark environment for 72 h. After the conditioning period, the weights of the swollen test pieces were recorded. The swollen test pieces were then dried in an oven at 70 °C for 15 min and allowed to cool at room temperature for another 15 min before the final weights were recorded. The Lorenzand Park's equation^{29–31} was applied in this study. The swelling index was calculated according to equation as follows:

$$Q_f/Q_g = ae^{-z} + b$$

where the subscripts f and g represent filled and gum vulcanisates, respectively; z is the ratio by weight of filler to hydrocarbon rubber in the vulcanisate; while a and b are constants. The higher the Q_f/Q_g value, the weaker the rubber-filler interaction becomes.

In this study, the weight of the toluene uptake per gram of hydrocarbon rubber (Q) was calculated based on equation as follows:

$$Q = [\text{Swollen} - \text{dried weight}] / [\text{initial weight} \times 100 / \text{formula weight}]$$

Viscoelastic properties of the samples were measured on a GABO EPLEXOR 500 N analyzer (GABO Instrument, Germany). The samples were trimmed to the following dimensions: 10 mm in length, 5 mm in width, and 1.0 mm in thickness. The properties were measured using tension clamp in the temperature range from −100 to 100 °C at a heating rate of 3 °C/min. The tests were carried out at a frequency of 11 Hz.

RESULTS AND DISCUSSION

SSBR Grafted *In Situ* via Solid State Method

A series of modified SSBR (MSSBR) were prepared in the process of mixing. The unmodified SSBR denoted as SSBR was given for comparison and the loadings of 3-mercaptopropionic acid for MSSBR were 0.2, 0.4, 0.8, and 1.6 phr. The changes of molecular structure were determined by ¹H-NMR spectroscopy. As can be seen in Table II, the molar content of 1,2-polybutadiene units decreased with the increasing of the loading of modification agents, while the molar content of 1,4-polybutadiene units remained unchanged, suggesting the grafting reaction occurred between the vinyl groups of 1,2-polybutadiene and 3-mercaptopropionic acid. The molar grafting ratio increased with the increasing of the loading of modification agents.

Molecular weight and molecular weight distribution before and after grafting are listed in Table III. It is obvious that the molecular weights keep almost constant after modification.

A comparison of SSBR and MSSBR (0.8 phr loading of 3-mercaptopropionic acid) is shown in Figure 1. It is clear that a chemical reaction between SSBR and 3-mercaptopropionic acid was identified by the appearance of the characteristic peak of 3-mercaptopropionic acid in modified SSBR after extraction in the ¹H-NMR spectrum, such as $-\text{SCH}_2-$ at a chemical shift of 2.75 ppm. The position of the peak is consistent with the report in the literature.²⁷

Table II. Molar Contents of Structural Units of SSBR and MSSBR

Sample code	Styrene unit (mol %)	1,2-Polybutadiene units (mol %)	1,4-Polybutadiene units (mol %)	Grafting ratio (mol %)
SSBR	15.5	47.3	36.7	0
MSSBR (0.2 phr)	15.3	46.3	36.7	1.7
MSSBR (0.4 phr)	15.2	46.0	36.7	1.9
MSSBR (0.8 phr)	15.4	45.4	36.7	2.6
MSSBR (1.6 phr)	15.3	44.5	36.7	3.1

Table III. Molecular Weight and Molecular Weight Distribution of SSBR and MSSBR

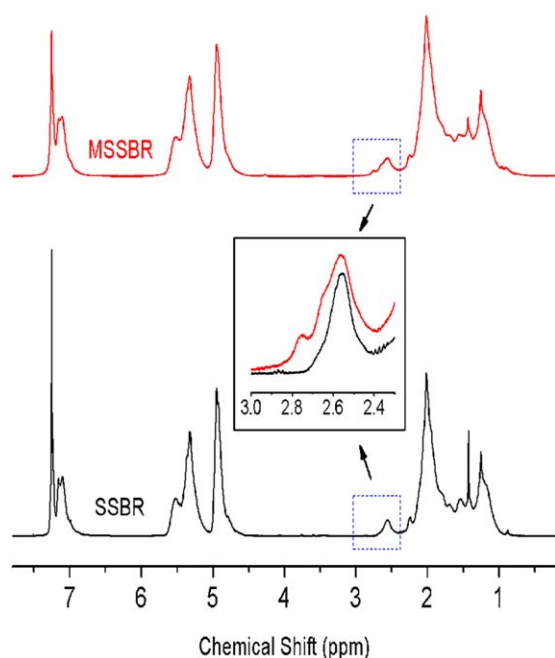
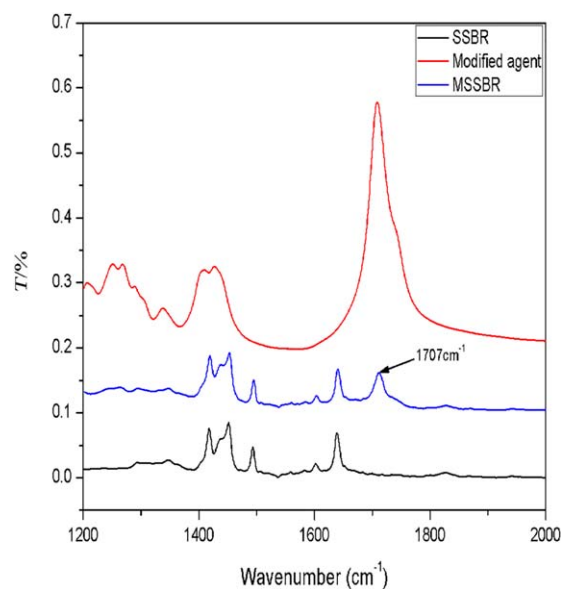
Sample code	$M_n \times 10^4$ (g mol ⁻¹)	$M_w \times 10^4$ (g mol ⁻¹)	M_w/M_n
SSBR	22.39	30.26	1.35
MSSBR (0.2 phr)	22.52	30.34	1.35
MSSBR (0.4 phr)	22.61	30.43	1.35
MSSBR (0.8 phr)	22.46	30.30	1.34
MSSBR (1.6 phr)	22.32	30.18	1.35

The detailed view of FTIR spectra from 1200 to 2000 cm⁻¹ shows the difference between SSBR and MSSBR (0.8 phr loading of 3-mercaptopropionic acid) in Figure 2. The strong absorbance at 1707 cm⁻¹ attributed to the —COOH vibration indicated that 3-mercaptopropionic acid was successfully grafted to SSBR chains in the solid state during the processing.

The reaction mechanism can be seen in Figure 3.

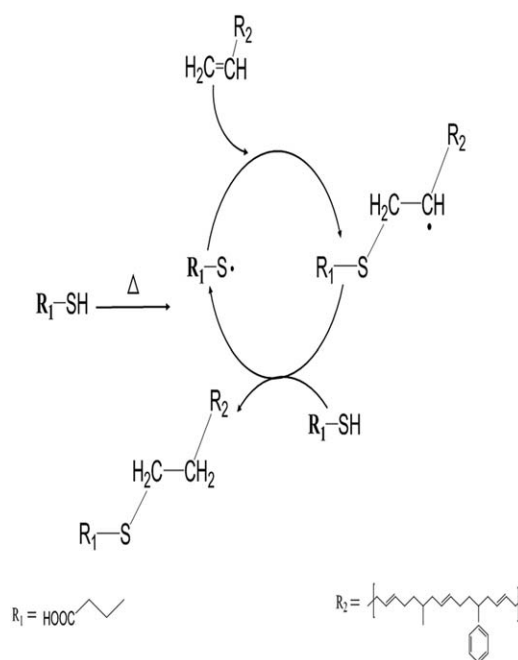
In our case, MSSBR was prepared in a solid state. An increasing loading of 3-mercaptopropionic acid resulted in the increasing of free radical concentration, partly due to a lower mixing efficiency. The more free radicals, the more crosslinked networks formed in the rubber matrix under a shearing action at high temperature,³² as shown in Figure 4.

The gel contents with increasing loading of 3-mercaptopropionic acid are plotted in Figure 5. With the increasing of the loading of 3-mercaptopropionic acid, the gel content increased gradually. When the loading of 3-mercaptopropionic acid was more than 0.8 phr, the gel content increased significantly by 462%.

**Figure 1.** ¹H-NMR Spectra of SSBR and MSSBR. [Color figure can be viewed at wileyonlinelibrary.com]**Figure 2.** FTIR spectra of SSBR and MSSBR. [Color figure can be viewed at wileyonlinelibrary.com]

Bound Rubber Content of SSBR/Silica Compounds of SPGiS

Bound rubber content of SSBR/silica compounds with the increasing loading of 3-mercaptopropionic acid is plotted in Table IV. The unmodified SSBR/silica compound denoted as sample 1 was given for comparison and the loading of 3-mercaptopropionic acid for MSSBR/silica compounds denoted as 2–5 were 0.2, 0.4, 0.8, and 1.6 phr. It shows that for samples 1–4 with the increasing loading of 3-mercaptopropionic acid, the bound rubber content increased gradually. The bound rubber is a layer of rubber that is adsorbed on the surface of filler, and the thickness is about a few nanometers. The rubber is

**Figure 3.** Grafting mechanism of SSBR and 3-mercaptopropionic acid.

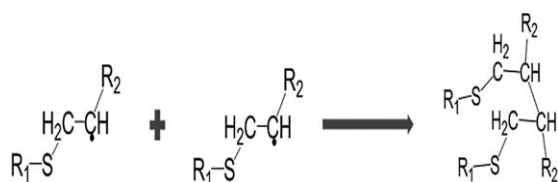


Figure 4. A formation of crosslink in the rubber matrix.

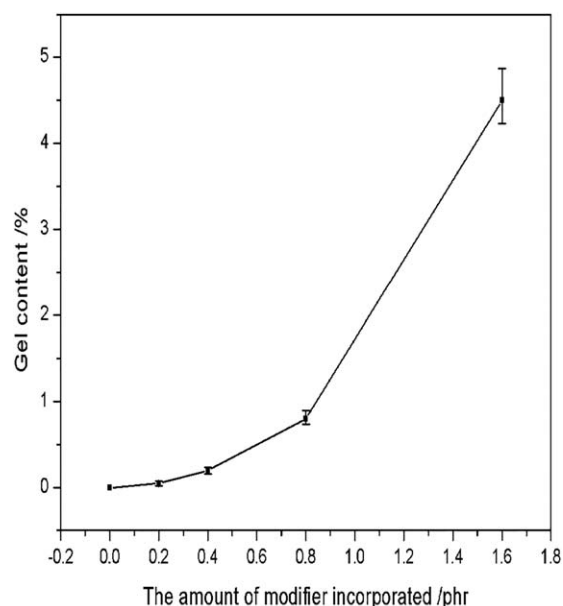


Figure 5. Gel content of SSBR and MSSBR.

firmly confined to the surface of filler, which is similar to the glass-like polymer.³³ Once the macromolecular chains polarized were strongly adsorbed on the surface of silica, it was hard to extract the macromolecules from the surface of filler by solvent toluene. Therefore, the higher the polarity degree of rubber, the stronger the interaction between rubber and silica, the higher the bound rubber content. Compared with the sample 1, the bound rubber content of sample 4 increased by 27.6%. As for sample 5, when the loading of 3-mercaptopropionic acid is more than 0.8 phr, the bound rubber content dropped. Because of the high gel content, the dispersion of silica in rubber was poor, which affected the interaction between silica and rubber.

Cure Characteristics of SSBR/Silica Compounds of SPGiS

Cure characteristics of SSBR/silica compounds with different polarity degree are presented in Table V. As for samples 1–4, with

Table IV. Bound Rubber Content of MSSBR/Silica Compounds

	Designation of the SSBR compounds based recipes				
	1	2	3	4	5
Bound rubber (%) ^a	52.6	55.7	58.7	67.1	54.9
Standard deviation	0.718	0.755	0.768	0.872	1.107

^aEach sample was tested five times in parallel and the value of bound rubber content was the average of the five results.

Table V. Cure Characteristics of MSSBR/Silica Compounds

Sample code	tc10 (min)	tc90 (min)	tc90 - tc10 (min)
1	2.9	29.3	26.4
2	2.8	26.3	23.5
3	2.6	25.9	23.3
4	2.0	19.5	17.5
5	2.7	27.4	24.7

Table VI. Torque Differences of MSSBR/Silica Compounds

Cure characteristics	Designation of the SSBR compounds based recipes					
	Unfilled	1	2	3	4	5
M_H (dN m)	4.95	30.30	31.23	31.59	32.5	32.80
M_L (dN m)	0.23	3.53	3.42	3.30	3.02	4.74
$M_H - M_L$ (dN m)	4.72	26.77	27.81	28.29	29.48	28.06

the increasing of polarity degree of SSBR, the scorch time and the cure time decreased gradually. Due to the high polarity of silica, silica not only interacts with zinc oxide (ZnO) during curing and formed silica bound zinc, which was unable to activate accelerators,^{34,35} but also affects the performance of accelerators through physical adsorption of hydrogen bonding which leads to the vulcanization retardation.^{36,37} Due to the high polarity of silica, silica could activate accelerators and ZnO at the same time during the processing which could retard the cure rate. After polarity modification of SSBR, the interaction between SSBR and silica was enhanced and accelerators and ZnO adsorbed was decreased, so the scorch time and the cure time were shortened.

As can be seen in Table V, compared with sample 1, the scorch time of sample 4 was decreased by 31.0% and the cure time of

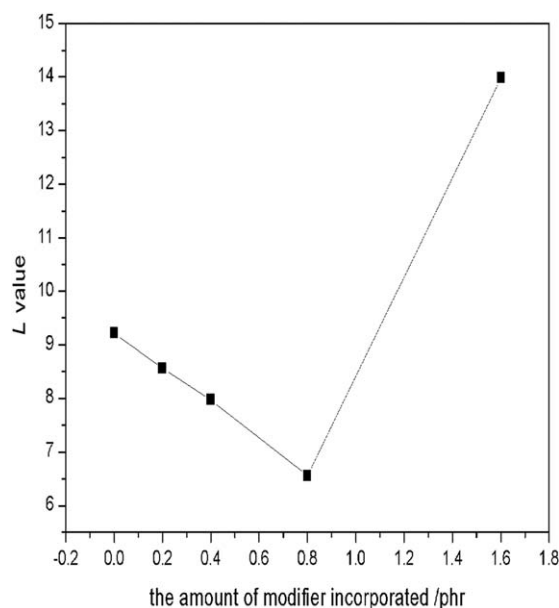


Figure 6. L values of MSSBR/silica compounds.

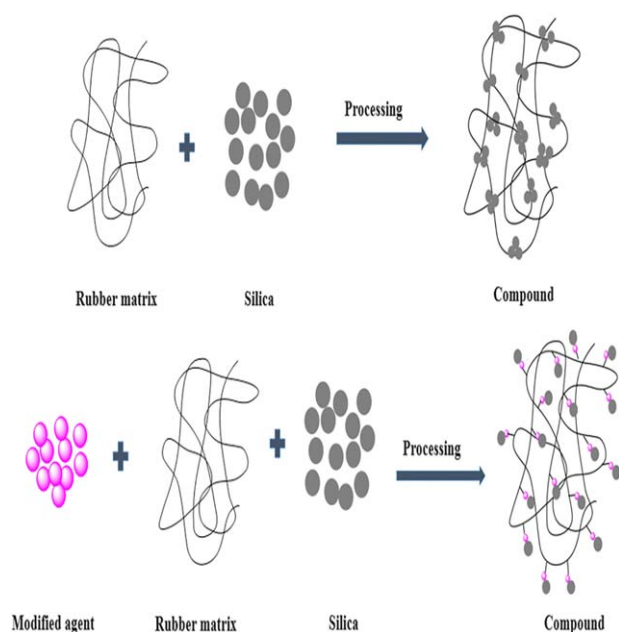


Figure 7. Sketch of the dispersion of silica in MSSBR and SSBR. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

sample 4 was decreased by 33.4%. However, sample 5 showed longer scorch time and longer cure time than sample 4. Mainly due to the strong crosslinking of rubber matrix, the interaction between rubber and silica was weakened, and the amount of accelerators and ZnO absorbed by silica was increased.

The torque difference ($M_H - M_L$) of SSBR/silica compounds with increasing loading of 3-mercaptopropionic acid is presented in Table VI. The minimum torque represented the filler-filler inter agglomeration,³⁸ and the value was used to measure the relative viscosity of a rubber compound.³⁹ The lower the value, the weaker the filler-filler interaction became. Due to the polarity of SSBR, the filler-filler interaction was reduced, Mooney viscosity was decreased and the processing performance of SSBR/silica compounds was improved.

In theory, the torque difference may be used as an indication of the crosslink density of a rubber compound. The greater the torque difference value, the higher the crosslink density. The total crosslink density is contributed by chemical crosslinks and physical crosslinks.^{40–45} In general, the carboxyl groups on the polarized SSBR chain and the hydroxyls on the surface of silica are connected by hydrogen bonds, that is to say, polarized SSBR and silica are connected by physical bond mainly, and the physical bonds contributed to the part of the crosslinks.

As for samples 1–4, with the increasing of polarity degree of SSBR, the torque difference increased gradually, as previously confirmed. The interaction between polarized SSBR and silica was improved, so the hydrophilicity of silica reduced, and then the amount of accelerators and ZnO adsorbed by silica decreased which accelerated the vulcanization reaction and promoted the formation of network structures. As for sample 5, compared with sample 4, the value of the torque difference dropped. Due to serious crosslinking of rubber matrix, the dispersion of silica became worse, and the interaction between SSBR and silica decreased.

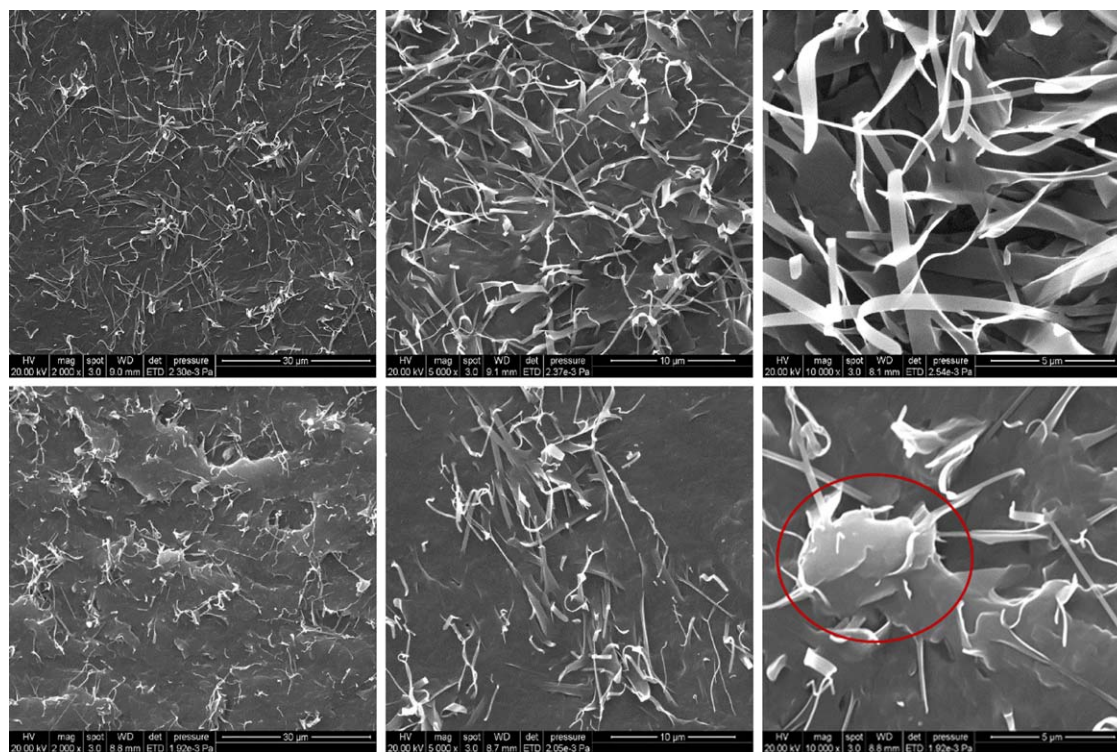


Figure 8. SEM micrographs of the tensile fractured surfaces of silica-filled vulcanizates at magnifications of 1000 $\times$, 5000 $\times$, 10,000 $\times$. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

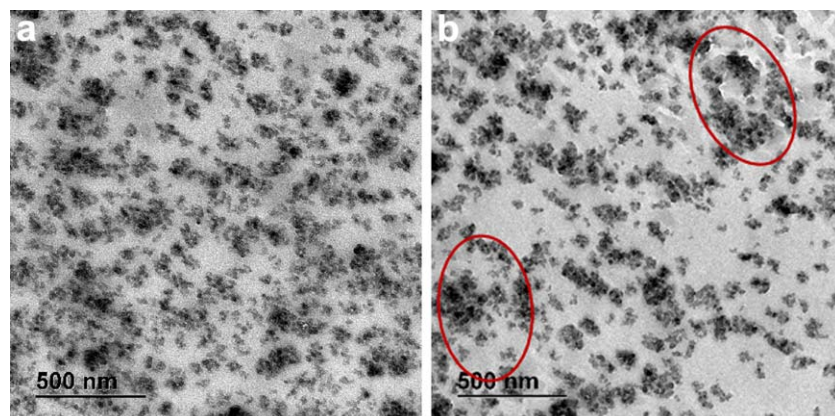


Figure 9. TEM micrographs of MSSBR/silica and SSBR/silica at magnifications of 12,000 $\times$. [Color figure can be viewed at wileyonlinelibrary.com]

Silica Dispersion of SSBR/Silica Compounds of SPGiS

The degree of silica dispersion was determined quantitatively by equation^{46,47}:

$$L = \eta_r - m_r$$

where $\eta_r = [M_{Lf}/M_{Lg}]$; $m_r = [M_{Hf}/M_{Hg}]$; where M_{Lf} and M_{Hf} were the minimum and the maximum torques of the filled compounds; and M_{Lg} and M_{Hg} were the minimum and the maximum torques of the unfilled/gum SSBR compound. From Table VI, $M_{Lg} = 0.23$, $M_{Hg} = 4.95$. The lower the value of L at a particular silica loading, the better the silica dispersion became in the SSBR. The value of L for the silica dispersion in SSBR is presented in Figure 6. As for samples 1–4, the value of L decreased with the increase of the polarity degree of SSBR. The introduction of polar groups on SSBR chains improved the compatibility and the interaction between rubber and silica, and thus improved the dispersion of silica in rubber. The value of L of sample 5, compared with sample 4, raised by 113.1%. Due to the excessive loading of 3-mercaptopropionic acid, the rubber matrix was crosslinked heavily. Therefore, the dispersion of silica in the matrix was blocked and large agglomerates of silica

were formed, because the silica particles saturated with hydroxyl groups had a strong tendency to interact with each other.

The sketch of the dispersion of silica in modified SSBR and unmodified SSBR can be seen in Figure 7. Via the SPGiS in the process, the polarity modifier became the bond between the filler and the rubber. Due to the enhancement of the polarity degree of SSBR, the filler network was not filler–filler association any more, but changed into a filler–rubber–filler linkage. This kind of interaction was much more stable than the one in unmodified SSBR/silica compounds, resulting in well-dispersed filler in rubber matrix.

SEM micrographs of the tensile fractured surfaces of silica-filled vulcanizes on MSSBR (0.8 phr loading of 3-mercaptopropionic acid) and SSBR are presented in Figure 8. As can be seen that filaments are evenly dispersed on the surface of SSBR/silica compounds of SPGiS, but not evenly dispersed on the surface of SSBR/silica compounds. When the magnification is 10,000 $\times$, agglomerations exit clearly. Due to the polarization of SSBR, the compatibility of SSBR and silica and the rubber–filler

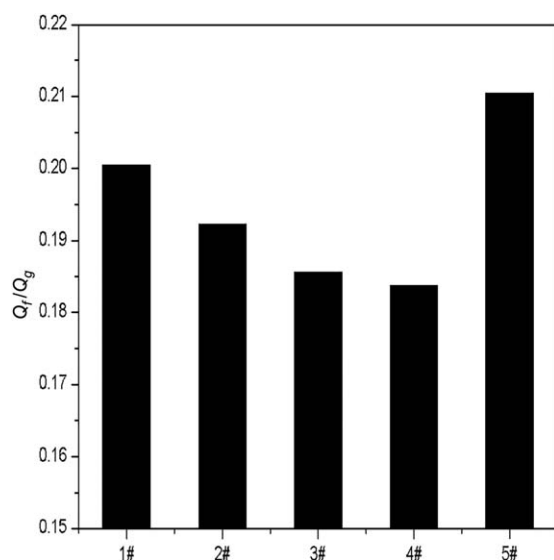


Figure 10. Q_f/Q_g values of MSSBR/silica compounds.

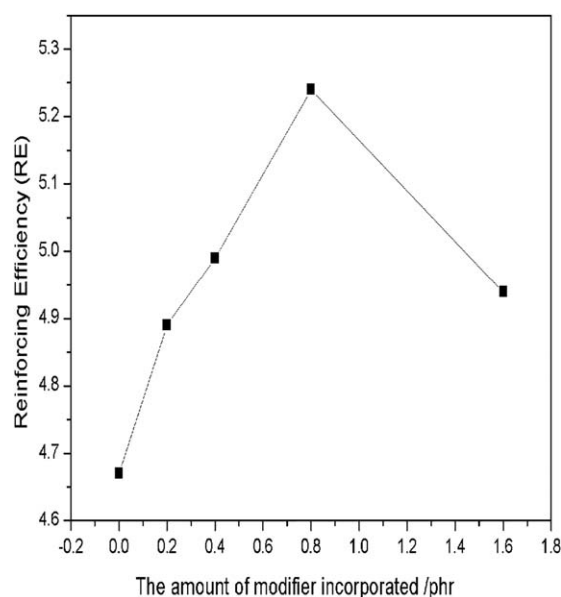


Figure 11. Reinforcing efficiency of MSSBR/silica compounds.

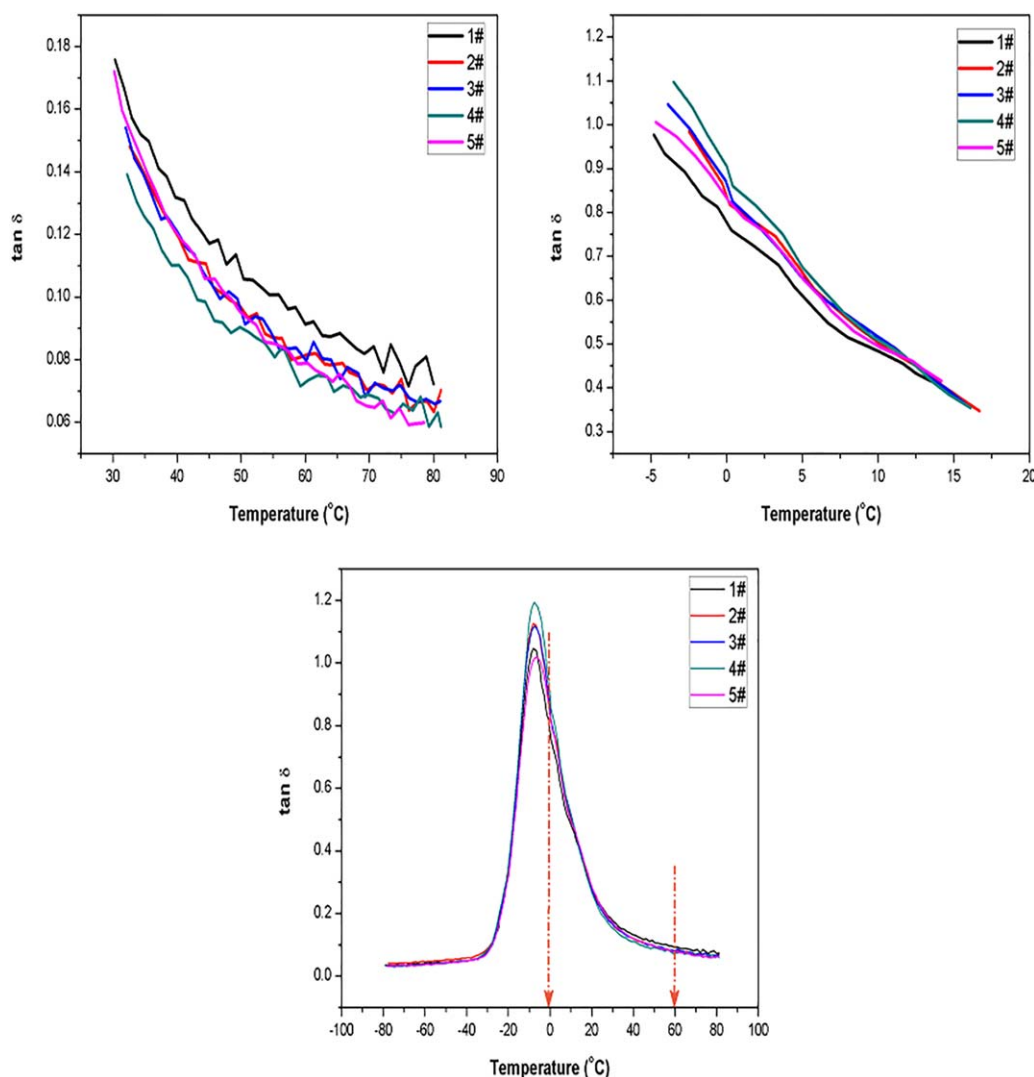


Figure 12. Loss factor versus temperature for MSSBR/silica compounds. [Color figure can be viewed at wileyonlinelibrary.com]

interaction were improved. When the vulcanizates were subjected to a certain tensile stress, not only the vulcanization network but also the network structure formed between the rubber matrix and silica would bear the stress together. Therefore, when the vulcanization network of matrix was destroyed, the network structure of the rubber matrix and silica was also destroyed, the sample was fractured. In Figure 8(a), the uniform filaments showed the better dispersion of silica in rubber matrix and the stronger interaction between rubber and silica. When tensile stress was applied on the vulcanizates, the rubber

molecules adsorbed on the surface of silica bore the tensile stress and at the same time part of rubber molecules were forced to pull out of the surface of silica and formed the filaments. As for SSBR/silica compounds, due to the poor dispersion of silica in rubber matrix and the weak interaction between rubber and silica, the filaments were not dispersed uniformly and agglomerations existed obviously [Figure 8(b)].

TEM micrographs support the results from SEM. TEM images of MSSBR/silica (0.8 phr loading of 3-mercaptopropionic acid) and SSBR/silica are displayed in Figure 9. It is evident that MSSBR/silica compound presents good dispersion of silica [Figure 9(a)]. However, as for SSBR/silica compound [Figure 9(b)], large silica agglomerates embed in SSBR/silica compound indicating relatively poor filler-rubber interactions.

Rubber-Filler Interactions of SSBR/Silica Compounds of SPGiS

The rubber-filler interaction depends on the degree of the filler dispersion in the rubber phase. Better filler dispersion resulted in stronger rubber-filler interactions. Rubber-filler interactions

Table VII. Dynamic Property Parameters for MSSBR/Silica Compounds

	Tan δ (0 °C)	Tan δ (60 °C)	T_g (°C)	Tan δ_{max}
1	0.7596	0.0913	-8.0	1.0460
2	0.8190	0.0818	-7.8	1.1257
3	0.8737	0.0797	-7.5	1.1174
4	0.9059	0.0734	-7.3	1.1940
5	0.8275	0.0788	-7.1	1.0176

of SSBR/silica compounds with increasing concentrations of 3-mercaptopropionic acid are presented in Figure 10. It shows that for samples 1–4 with the increase of the content of 3-mercaptopropionic acid, the value of Q_f/Q_g decreased gradually. Due to the polarity of SSBR, the wetting dispersion of silica was improved in the rubber matrix; hence, the compatibility of SSBR and silica and the rubber-filler interaction was improved. This explanation is in agreement with Figure 6. As for sample 5, because of the crosslinking of SSBR in the modification process, the dispersion of silica was poor and rubber-filler interaction was weak.

Reinforcing Efficiency of Silica in SSBR/Silica Compounds of SPGiS

The degree of reinforcement provided by the filler was calculated through its reinforcing efficiency (RE), which in its simplest form is given by equation,⁴⁷

$$RE = [(M_H - M_L)_f - (M_H - M_L)_g] / (M_H - M_L)_g$$
 in which: where $(M_H - M_L)_f$ is the difference in torque value of filled compound and $(M_H - M_L)_g$ is the difference in torque value of unfilled/gum compound.

High RE meant a high rubber-filler interaction, which was influenced by the degree of filler dispersion. Enhanced filler dispersion provided a greater surface area for rubber-filler interactions. Reinforcing efficiency of silica with increasing loading of 3-mercaptopropionic acid is presented in Figure 11. It shows that with the increasing of the content of 3-mercaptopropionic acid, the value of RE increased gradually. The decrease of RE after 0.8 phr loading was probably due to the excessive loading of 3-mercaptopropionic acid. That was attributed to the polarity modifier playing a role as a bridge between rubber and filler, which contributed to better filler dispersion and stronger rubber-filler interaction. As for sample 5, the existence of large amount of gel led to poor dispersion of silica and worse reinforcing efficiency.

Viscoelastic Behaviors of SSBR/Silica Compounds of SPGiS

It is well known that temperature has special significance for tread tires.^{48,49} A higher hysteresis at 0 °C and lower hysteresis at 60 °C would be favorable for higher wet skid resistance and lower rolling resistance, respectively. Figure 12 presents $\tan \delta$ curves obtained for SSBR/silica compounds of SPGiS. T_g , taken as the temperature of maximum $\tan \delta$, together with maximum $\tan \delta$ ($\tan \delta_{\max}$) are collected in Table VII. From the DMA measurements, we detected the changes in relaxation behaviors with the increasing loading of the modification agent. Due to the strong interactions between rubber and filler, the intensity of T_g relaxation increased. Correspondingly, the better the filler dispersion, the higher the value of $\tan \delta_{\max}$.⁵⁰ In the glass transition zone, the main source of energy dissipation was from the polymer matrix, while the filler network could not be easily broken. The hysteresis loss would be significantly attenuated by filler agglomeration because of the reduction of the effective volume of the polymer matrix.⁵¹ With the increase of the polarity degree of SSBR, the dispersion of silica was improved, the effective volume of the polymer matrix was increased, and the hysteresis at 0 °C was increased accordingly. In the high temperature zone (20–70 °C), the hysteresis decreased with the increase

of the polarity degree of SSBR. Due to the polarity of SSBR, the rubber and filler were combined tightly by chemical bonds and the interaction between the rubber and the filler was strengthened, followed by reduced internal friction between filler and filler. Take sample 1 as a reference, the wet skid resistance of sample 4 was improved by 19.3% and the rolling resistance was decreased by 19.6%. As for sample 5, due to the serious crosslinking of rubber matrix, the dispersion of filler was poor and the interaction between rubber and filler was weak. Therefore, the dynamic properties of sample 5 became worse.

CONCLUSIONS

From this study, the following conclusions were drawn:

1. The feasibility of solid-phase grafting *in situ* (SPGiS) was proved by NMR and FTIR.
2. A series of modified SSBR with different polarity degrees were prepared through grafting 3-mercaptopropionic acid onto SSBR chains in the process of mixing.
3. With the increasing of polarity degree of SSBR, the scorch time and vulcanization time of MSSBR/silica compounds decreased gradually.
4. With the increasing of polarity degree of SSBR, filler dispersion and rubber-filler interactions and reinforcing efficiency were improved.
5. SEM and TEM showed that the size of silica aggregates was reduced and the filler dispersion was improved significantly in the polarized SSBR/silica compounds.
6. When the loading of 3-mercaptopropionic acid was 0.8 phr, the wet skid resistance of SSBR/compound was improved by 19.3% and the rolling resistance was decreased by 19.6%.
7. 0.8 phr loading of 3-mercaptopropionic acid was the optimal loading to improve the properties of SSBR/silica compounds. Overloading of 3-mercaptopropionic acid was counterproductive to the material performance.

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